

The Price of Portability: How Interstate Licensing Transforms Mental Health Markets

Cici McNamara[†]

Abstract

Interstate occupational licensure compacts are an increasingly popular way of addressing provider shortages in mental healthcare markets. I study the effects of licensure compacts on the size and composition of the psychologist workforce. I develop an empirical model of entry for psychologists differentiated by whether they provide in-person or virtual care. Results show that the Psychology Interjurisdictional Compact increased overall supply by 5.71 psychologists per market, with both in-person and virtual supply rising. The direct effect of compact membership on in-person supply is large, but is partially offset in equilibrium by competitive pressure from virtual entrants, reducing the net in-person gain to 55%. This indirect crowding-out falls disproportionately on markets already well-supplied with in-person providers. Geographic license portability unambiguously expands psychologist supply, with distributional benefits concentrated in underserved markets.

Keywords: Occupational licensing, Interstate compact, Mental health, Labor supply

JEL codes: J44, J21, I18, L10

[†]School of Economics, Georgia Institute of Technology, Atlanta, Georgia, USA. cmcnamara30@gatech.edu. I am grateful to the Institute for Humane Studies for their support; to Campbell Clark for her excellent research assistance; to Cecilia Diaz-Campo, James Flynn, Jason Lindo, Conor Ryan, and Christopher Sullivan for their useful comments; and to participants of the Public Choice Society, Midwest Economics Association, and International Industrial Organization annual conferences as well as the Southeastern Micro Labor Workshop for their feedback. All errors are my own.

1 Introduction

Markets for mental health care have evolved dramatically in recent years, and especially since the onset of the COVID-19 pandemic. Evidence suggests that the supply of mental health care providers is not sufficient to meet new levels of demand. As of 2025, over 137 million individuals reside in a mental health workforce shortage area ([KFF, 2025](#)). Shortages are especially acute for providers of psychotherapy, which is often referred to as talk therapy or colloquially as “therapy.” The effects of this shortage are being felt by providers and patients: in 2023, 56% of psychologists reported having no openings for new patients, and among those who kept a waitlist, the average reported time spent on a waitlist for an initial appointment was over three months ([APA, 2023](#)). This is particularly concerning in light of research finding that longer wait times for mental health visits result in higher mortality rates among individuals experiencing mental health emergencies ([Costantini, 2025](#)).

There are several constraints on psychotherapist supply that may be contributing to the shortage. Some of the most pertinent barriers identified by stakeholders include low reimbursements for behavioral health services; burnout, particularly since the start of the pandemic; and the high costs of occupational licensure ([GAO, 2022](#); [Thom and Norris, 2023](#)). In this paper, I examine how occupational licensure regulations affect the size and composition of the psychotherapist workforce.

To be licensed and practice independently, therapists must generally earn a graduate degree, complete a certain number of hours of supervised practice, pass required exams, and pay fees. The specifics of these licensure requirements vary across professions and states. Therapists must also apply for and maintain licensure separately in each state in which they practice. Licensure costs may prevent therapists from seeking licensure in multiple states, which can disrupt continuity of care for patients who move to a state in which their therapist is not licensed. Low rates of cross-state licensure have become particularly constraining with the widespread adoption of teletherapy, use of which has stabilized to 10 times pre-pandemic levels, as well as with increases in state-to-state migration in the U.S. over the last decade ([Cantor et al., 2023](#); [Ismail, 2023](#)).

In response to these trends, many states and professional associations have begun to form licensure compacts. These compacts facilitate the practice of psychotherapy across state lines by making it so that licensure in one compact member state implies licensure in all other compact member states. The Psychology Interjurisdictional Compact (PSYPACT), implemented in 2019, allows psychologists to practice teletherapy in any state that is a member of the compact ([PSYPACT, 2025](#)). The Counseling Compact does the same for licensed professional counselors ([Counseling Compact, 2025](#)). By enabling providers who offer in-person care to expand their patient base through virtual channels, PSYPACT may increase the overall profitability of entry and increase the availability of in-person care. Alternatively, licensure compacts such as these potentially increase the profitability of being a virtual provider relative to an in-person one by reducing the costs of entering out-of-state markets, no matter their geographic proximity to the provider. To the extent that virtual and in-person therapy are substitutes for each other, increased entry of virtual providers may also put competitive pressure on in-person ones, resulting in a further decline in in-person provider supply. The overall impact on in-person supply is theoretically ambiguous.

I examine how occupational licensure requirements impact the structure of talk therapy markets, quantifying how licensure compacts differentially affect entry across differentiated providers. I focus on the entry decisions of psychologists, who diagnose and treat mental health conditions through psychological testing and talk therapy. I identify the practice locations and modalities of psychologists using the universe of profiles maintained by Psychology Today, the leading online directory that mental health professionals use to advertise their services and connect with potential clients. Psychologists on Psychology Today can choose to advertise their services in a pre-specified number of markets. Entry in geographically distant markets is made possible by teletherapy and the ability to enter as an online-only provider.

I categorize psychologists into three types: those who offer in-person services and are local to the markets they have entered; those who are local but choose to practice online only; and those who are physically located in another state and so only provide online therapy. I construct counts of the number of psychologists of each type practicing in each market and develop an empirical

entry model that relates entry to own- and cross-type competition as well as to whether the market is in a PSYPACT member state. The model allows for correlation in unobserved profit shocks across types, using an affordability index that measures the cost of living borne by local providers to separately identify the correlation in shocks from the cross-type competition effects. Because PSYPACT membership is likely correlated with unobservable determinants of profits, I employ a control function approach that uses the number of licensure compacts a state had joined prior to PSYPACT's implementation as an instrument for membership.

My estimates indicate that virtual entry does put competitive pressure on in-person psychologists: markets with at least one virtual competitor have 18.4% fewer in-person providers on average. I additionally find that PSYPACT membership significantly increases both out-of-state psychologist entry (27.5%) and in-person psychologist entry (16.3%), with no significant effect on local online providers. To investigate the drivers of psychologist shortages, I apply the entry threshold methodology of [Bresnahan and Reiss \(1991\)](#) to characterize the minimum market size required for the first in-person psychologist to enter. In PSYPACT member markets, this threshold falls well below the 6,000-person criterion used to designate Mental Health Professional Shortage Areas, while the corresponding threshold in non-member markets far exceeds it. Consistent with the competition results above, the presence of virtual competitors modestly raises in-person first-entry thresholds in both groups. These partial equilibrium estimates suggest that PSYPACT's direct benefits to in-person access outweigh its indirect costs via virtual competition.

I use the model estimates to compute the equilibrium effects of compact membership taking into account this indirect competitive effect. I find that compact membership increases virtual provider supply by 0.37 (20%), and that this increase in competition offsets approximately 45% of the direct in-person gains, yielding a net increase of 5.34 in-person psychologists per market (55%). The compact therefore expands overall psychologist supply by 5.71 providers per market. Importantly, this indirect crowding-out falls disproportionately on markets relatively well-supplied with in-person providers at baseline, so that the areas most in need of care benefit more from compact implementation.

These patterns have implications for how we think about standard measures of supply, which are usually geographically based, as well as for the likely effects of such licensure policies on patients. Despite recent research finding virtual talk therapy to be equally as effective as that provided in-person, my results suggest that modality of provision meaningfully differentiates providers ([Pescatello et al., 2021](#); [Andrews et al., 2018](#)). This differentiation lessens the extent of competition across modalities and ensures that expansion of virtual care need not come at the expense of in-person provider supply.

This paper contributes to a growing literature on the impacts of occupational licensure requirements for professionals who market their services via digital platforms ([Blair and Fisher, 2022](#); [Farronato et al., 2024](#); [Deyo, 2022](#)), which has found that higher occupational licensure requirements result in less competition and higher prices without improving consumer satisfaction. Using the static entry framework of [Bresnahan and Reiss \(1990, 1991\)](#), I likewise find that reductions in cross-state licensure requirements result in an overall increase in competition. I additionally show that these competitive effects have a meaningful impact on the composition of psychologist labor supply.

This paper also contributes to the literature investigating the determinants of entry into U.S. healthcare markets ([Bresnahan and Reiss, 1991](#); [Magnolfi et al., 2024](#); [Abraham et al., 2007](#); [Cohen et al., 2013](#)). [Goetz \(2023\)](#) is the only other work to consider how the entry of virtual competitors impacts psychotherapy market outcomes. The author finds that price-elastic consumers substitute to virtual therapists following entry, resulting in higher post-entry prices for in-person incumbents. Relative to [Goetz \(2023\)](#), this paper endogenizes the therapist entry and modality decisions on the extensive margin, but is unable to assess intensive margin effects on price or utilization.

Finally, this paper contributes to the literature on occupational licensure and healthcare access. Research on licensure compacts has focused on their impact on the geographic mobility of providers, finding mixed results ([DePasquale and Stange, 2016](#); [Shakya et al., 2022](#); [Deyo et al., 2016](#)). While compacts expand the geographic scope of practice, scope of practice laws expand the set of services permitted under licensure and have been found to increase access to care ([Alexan-](#)

der and Schnell, 2019; Patel et al., 2019). I extend this literature to the market for mental health services and show that licensure compacts can increase access to all modalities of mental health care.

I proceed as follows: Section 2 describes the data; Section 3 provides descriptive evidence of how licensure policies affect psychologist entry patterns; Section 4 presents the empirical entry model; Section 5 presents estimation results; Section 6 quantifies the equilibrium effects of licensure compacts on psychologist supply; and Section 7 concludes.

2 Setting and data

2.1 Data on number of psychologists

For information on the size and composition of the psychotherapy workforce, I use data from an online directory of therapists maintained by Psychology Today, the leading website that mental health professionals use to advertise their services and connect with potential clients. A Psychology Today therapist directory listing costs \$29.95 per month and includes an active listing, client messaging, and a telehealth platform. The directory allows for filtering by geography and professional type, the latter of which facilitates my focus on psychologists. This data was scraped in January 2025 and is a snapshot of the website at that time.

The Psychology Today data provides multiple advantages over alternative datasets. First, many therapists are self-employed: survey data from the American Psychological Association indicates that 43% of psychologists engaged in individual solo practice in 2021 (APA, 2021). For this reason, many psychotherapists will be excluded from datasets that are limited to workers who are not self-employed or are based in physician offices, such as the employment estimates of the Bureau of Labor Statistics or the National Ambulatory Medical Care Survey. Second, over one-third of psychologists do not accept insurance and so will be excluded from claims databases (APA, 2024).

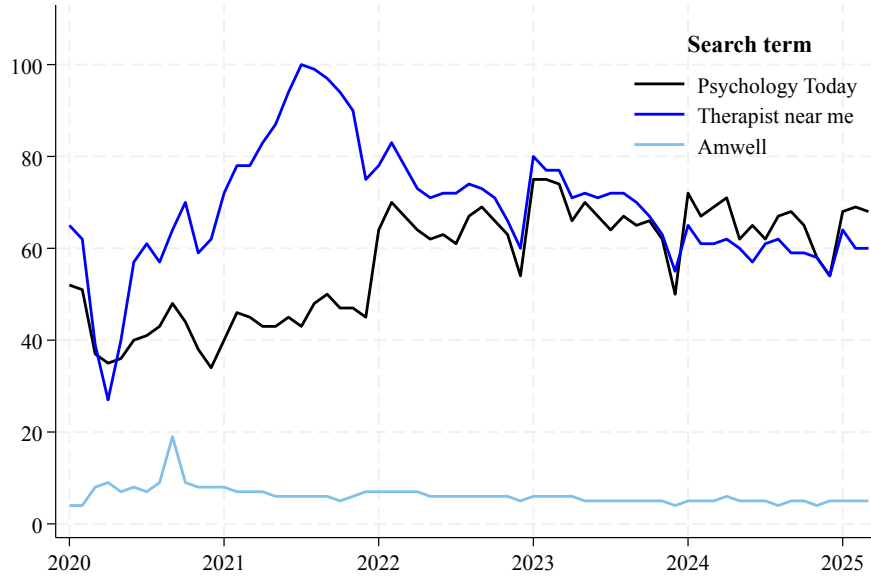
As an example, Figure A1 provides a comparison of psychologist counts in the Psychology

Today data against those in the Bureau of Labor Statistics Occupational Employment and Wage Statistics (OEWS). The light and dark blue dots correspond to natural logarithm of each state's number of psychologists as given by the Psychology Today data and OEWS, respectively. The population-weighted difference in these log counts averages 0.32, and the correlation in the level count of psychologists across the two data sources is 0.97. These patterns together suggest meaningful and widespread undercounting of psychologists in the BLS relative to Psychology Today. While it is possible that some psychologists maintain active practices through referrals or institutional affiliations without a directory listing, the high correlation between Psychology Today and BLS counts across states suggests that any undercounting of psychologists who maintain practices without a directory listing is unlikely to vary substantially across markets in ways that would bias the estimated effects.

Psychology Today contains both a breadth and depth of information on providers, including information on their demographic characteristics, areas of specialization, accepted insurance plans, and educational background, making it superior to more limited directories, including the online provider tools of individual insurers. Figure 1 shows the trend in Google searches for terms one might use to find a psychologist over the five years preceding the sample period. As of mid-2025, searches for “Psychology Today” were higher than those for competing online directories of therapists as well as for the very general search term “therapist near me,” providing confidence that provider counts constructed with the Psychology Today data are the best available approximation of psychologist supply.

I categorize psychologists on the platform into one of three types, which are determined by the modalities they offer, their physical location, and the geographies in which they choose to advertise their services. Psychologists can choose from one of three modality categorizations: online only, in-person only, or both in-person and online. Each psychologist's listing is automatically included in the search results for the area corresponding to their physical location if they offer in-person services. In-person providers may additionally select up to two zip codes in which to advertise their services, while online-only providers select all three zip codes in which they advertise. I

Figure 1: Trend in Google search terms for psychologists



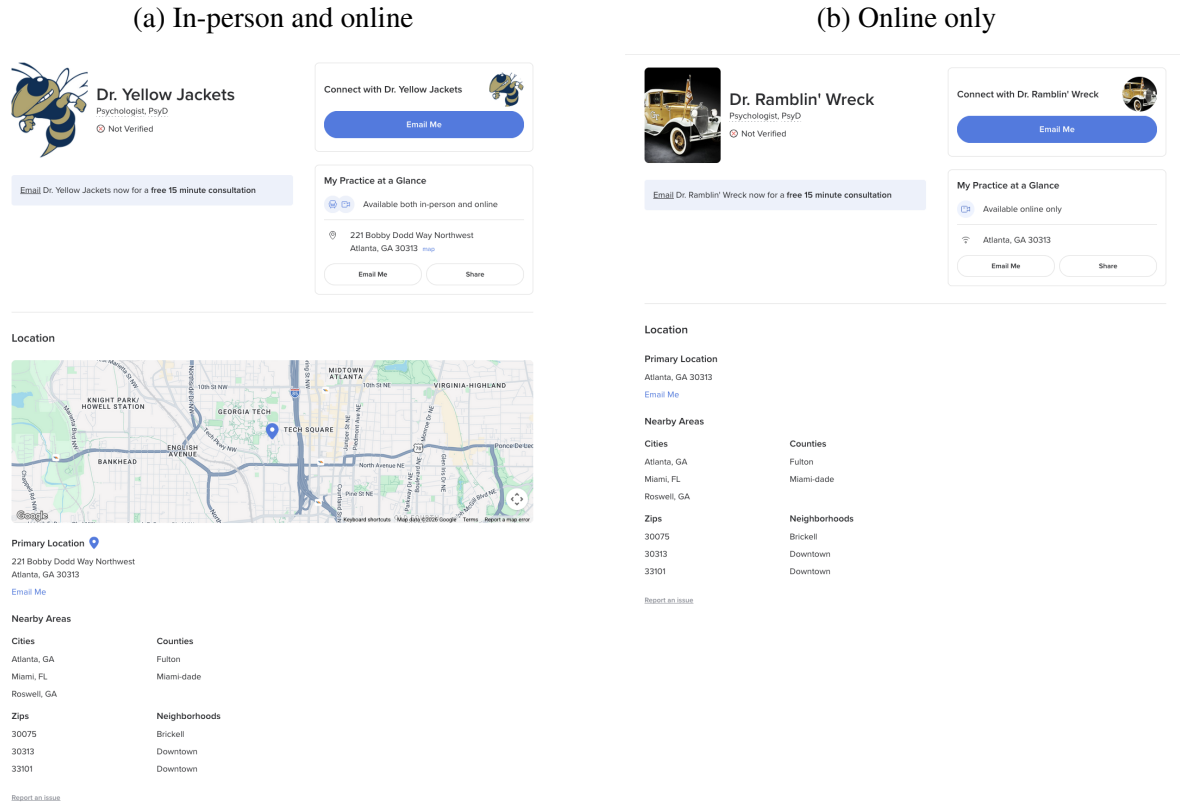
Notes: Data come from Google trends: trends.google.com/trends/.

define an in-person psychologist as one who offers in-person services and is physically located in the same state as the market in which their services are being advertised, so that in-person provision is actually possible. This definition includes psychologists who offer both in-person and online services, as 87% of psychologists who offer therapy in-person also offer it online. As a result, entry as an in-person provider nearly always entails simultaneous entry as a virtual one; the in-person categorization reflects the fact that consumers in a market with an “in-person” provider *have the option* of receiving care in person, rather than that the provider delivers care exclusively through this modality.

I define out-of-state therapists as those who are located in a different state from that in which their services are being advertised and so offer online services only. Lastly, I define local online therapists as those who are physically located in the same state as the market in which their services are being advertised but offer online services only. These categorizations are mutually exclusive and collectively exhaustive partitions of the directory listings.

To fix ideas, Figure 2 provides example profiles for two psychologists who operate in the same

Figure 2: Example Psychology Today profiles



Notes: Screenshots are of profiles made on Psychology Today for illustrative purposes only.

markets but offer different modalities. Dr. Yellow Jackets will be categorized as an in-person provider for all of the Georgia markets that they choose to list on their profile, but as an out-of-state provider in Miami. Note again that this categorization is based on the extensive margin of provider choice and reflects the fact that consumers in the Georgia markets have the option to receive in-person therapy from Dr. Yellow Jackets, while this is not possible for consumers located in Miami. Dr. Ramblin' Wreck will be categorized as a local online provider for all of their Georgia markets and as an out-of-state provider in Miami. The Psychology Today platform solicits and validates licensure information from all providers; in this example, we can therefore assume that both providers are licensed to practice psychology in both Georgia and Florida, or that Georgia and Florida are both members of an interstate licensure compact.¹

¹To provide evidence that PSYPACT is a driver of out-of-state entry, I scraped all 15,557 entries in the PSYPACT directory (<https://directory.psypact.gov/>) and matched them to Psychology Today providers by name using a strict name-matching procedure. Out-of-state providers match a PSYPACT directory entry at twice the rate of

This psychologist-type categorization has two main advantages. First, licensure compacts reduce the barriers to across-state entry only. Although across-state entry necessarily implies online provision, online provision does not necessarily imply out-of-state entry. Drawing this distinction between out-of-state and local providers allows me to empirically test whether licensure compacts differentially impact supply of these two types of providers. Second, the welfare implications of licensure compacts are likely very different depending on the extent to which they impact the geographic distribution of providers. The replacement of in-person psychologists with out-of-state ones reallocates providers across space in a way that the transition from in-person to local online provider does not, and the former may have much more serious welfare implications for those for whom virtual care is inadequate.

2.2 Market definition

Psychology Today prompts users to provide their location to begin a search for both in-person and virtual providers. This means that while patients could technically choose any virtual provider regardless of the provider's location, they are presented a choice set of virtual (and in-person) providers who have chosen to have their services advertised in the patient's market. Other popular online directories of medical providers also offer location-based recommendations for virtual providers. A shared market definition for in-person and virtual therapists is therefore appropriate even if entry costs for geographically distant markets are lower for the latter.

I follow [Cohen et al. \(2013\)](#) in defining markets as counties. As [Figure 2](#) shows, providers select zip codes, which the platform aggregates to counties and cities that patients use as search terms. Psychotherapy is typically used to treat chronic mental health conditions rather than acute care conditions, so a larger market definition than that used in studies of entry of acute care providers is warranted ([Magnolfi et al., 2024](#)). The role of counties as the fundamental unit of geographic search on the platform and their prior use in defining markets for outpatient care providers makes local ones, consistent with PSYPACT being a causal determinant of out-of-state entry.

it a natural market definition for this analysis.

Table 1: County characteristics by number of in-person psychologists

	Number of in-person psychologists					
	0	1	2	3	4	≥ 5
Competitor counts						
Out-of-state	0.0 (0.3)	0.1 (0.6)	0.3 (0.8)	0.4 (1.0)	0.5 (1.0)	10.5 (36.0)
Local online	0.1 (0.3)	0.4 (0.8)	0.5 (0.9)	0.7 (1.1)	1.2 (1.5)	34.5 (120)
Policy environment						
PSYPACT	86.2%	88.8%	87.6%	82.6%	87.9%	80.9%
Pre-PSYPACT compact count	3.5 (1.1)	3.3 (1.3)	3.2 (1.4)	3.0 (1.6)	3.3 (1.4)	2.8 (1.6)
Hours (thous.)	4.1 (1.0)	4.2 (1.0)	4.1 (1.0)	4.1 (0.8)	4.1 (1.1)	4.1 (0.9)
Fees (\$h)	2.1 (1.0)	2.0 (1.1)	2.2 (1.1)	2.3 (1.2)	2.0 (1.2)	2.3 (1.2)
MHPSA designation rate	60.3%	49.4%	43.1%	44.6%	33.3%	23.8%
Socio-demographics						
Population (10k)	2.1 (2.2)	4.3 (3.2)	6.0 (4.0)	7.8 (6.3)	9.6 (5.7)	42.5 (67.1)
Median income (\$10k)	6.0 (1.3)	6.6 (1.3)	6.8 (1.5)	7.3 (1.5)	7.2 (1.2)	8.4 (2.1)
Share insured	90.1 (5.4)	90.7 (4.6)	91.1 (4.5)	92.5 (3.5)	91.0 (4.3)	92.5 (3.7)
Share unemployed	4.6 (2.9)	4.6 (1.8)	4.8 (2.0)	4.4 (1.5)	4.5 (1.7)	4.6 (1.5)
Share Black	8.3 (15.1)	8.0 (13.2)	10.1 (15.6)	7.5 (11.2)	8.9 (11.1)	10.4 (11.6)
Share Hispanic	8.9 (14.1)	10.0 (12.9)	9.5 (13.6)	9.0 (10.2)	9.0 (6.8)	14.3 (14.0)
Share over 65	21.0 (4.9)	20.0 (4.5)	19.2 (4.2)	18.9 (4.8)	19.0 (4.1)	17.4 (4.8)
Share $\leq$ high school	36.6 (6.4)	34.5 (6.1)	33.4 (6.1)	31.7 (6.7)	30.0 (5.9)	25.3 (6.1)
Share rural	38.4 (40.1)	20.2 (28.4)	13.5 (23.3)	14.0 (24.5)	9.1 (21.3)	5.1 (13.4)
Share with broadband	81.9 (6.4)	84.8 (5.1)	85.8 (5.2)	87.8 (4.1)	88.0 (2.8)	90.6 (3.2)
Affordability index	100.3 (1.0)	100.0 (0.8)	99.9 (0.9)	99.6 (0.7)	99.6 (0.6)	99.2 (0.7)
Observations	1,904	340	137	92	66	602
Share of all observations	60.6	10.8	4.4	2.9	2.1	19.2

Notes: In-person psychologist and competitor counts come from Psychology Today. PSYPACT is a binary indicator for participation in the Psychology Interjurisdictional Compact. The affordability index is computed using price parity and income data from the Bureau of Economic Analysis ([BEA, 2024](#)). Pre-PSYPACT interstate licensure compacts include the Nurse Licensure Compact (2000), EMS Compact (2017), Physical Therapy Compact (2017), and Interstate Medical Licensure Compact (2017). All socio-demographic characteristics come from the American Community Survey ([Census Bureau, 2025](#)). MHPSA stands for Mental Health Professional Shortage Area.

Table 1 provides summary statistics for all counties using data from Psychology Today and area-level characteristics from a variety of sources. Over half of counties have no in-person psychologists. The number of virtual psychologists trends monotonically with the number of in-person psychologists, which is potentially reflective of correlation in the profitability of entry across types. In general, there are more in-person psychologists in areas that are more populous, higher income,

more racially diverse, younger, less rural, more educated, and have greater rates of broadband internet adoption. The Mental Health Professional Shortage Area (MHPSA) designation rate—a federal measure of inadequate mental health provider supply—falls from 60% in counties with no in-person psychologists to 24% in those with five or more, consistent with the provider counts capturing meaningful variation in access to care.

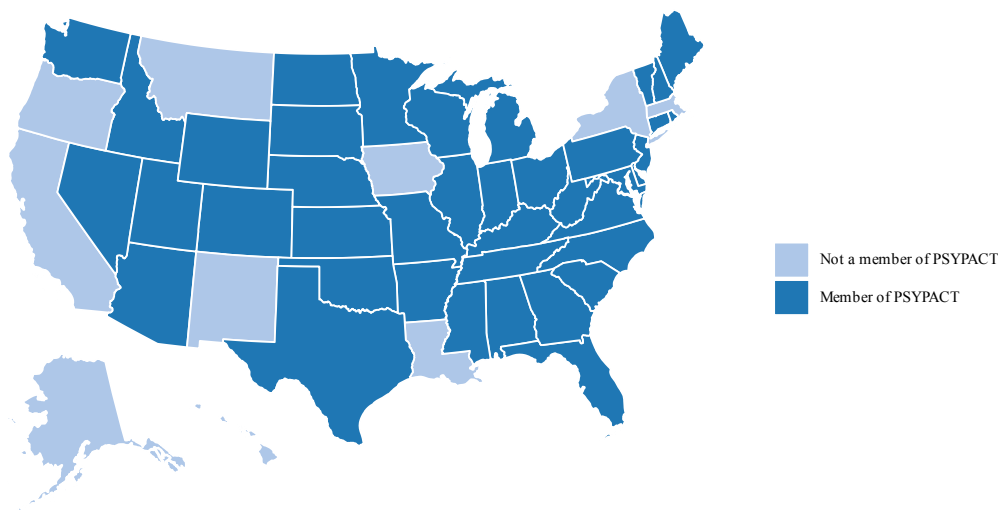
2.3 The Psychology Interjurisdictional Compact

Interstate licensure compacts are agreements between states to broaden the geographic scope of licensure to include all compact member states. Licensure compacts are not new to healthcare: since its inception in 2000, 40 states have joined the Nursing Licensure Compact (NLC), which permits registered nurses living in member states to practice across state lines. Research has found that the NLC had little effect on the labor market supply or mobility of nurses ([DePasquale and Stange, 2016](#)). Research on the Interstate Medical Licensure Compact, implemented in 2017, finds that it resulted in a 3% increase in out-of-state practices for physicians whose home state participated in the compact ([Deyo et al., 2016](#)). The Psychology Interjurisdictional Compact (PSYPACT) is an interstate licensure compact for psychologists that was implemented in 2019. Unlike previous licensure compacts that sought to promote physical geographic mobility among healthcare providers, the explicit aim of PSYPACT was to expand access to psychotherapy by “allow[ing] for the delivery of psychological services via telecommunications from providers to patients in separate states” ([PSYPACT, 2023](#)). It is the first licensure compact for mental health professionals. [Figure 3](#) depicts the 40 U.S. states that have joined PSYPACT and the 10 that have not as of 2025.

To practice teletherapy under PSYPACT, psychologists must hold an active, unrestricted license in their home state that is a PSYPACT member. They must obtain both an E.Passport certificate from the Association of State and Provincial Psychology Boards, which verifies their credentials and good standing, and an Authority to Practice Interjurisdictional Telepsychology from the PSYPACT Commission. The application therefore involves background checks and verification of existing licensure, but no additional licensure. In the absence of a compact, psychologists seeking

to provide teletherapy in another state would have to become licensed in that state. The costs of additional licensure are non-trivial, as supervised hours requirements, fees, and renewal frequencies vary substantially across states. For example, the supervised hours requirement for psychologists varies from 1,500 to 4,000 hours, and initial licensure fees vary from \$50 in Idaho to \$600 in Nevada (NPTC, 2025; ASPPB, 2025). Licensure compacts may affect the profitability of entering the psychology profession, practicing virtually relative to in-person, and the level of competition between different types of providers. I model each of these effects in the following section.

Figure 3: Psychology Interjurisdictional Compact membership



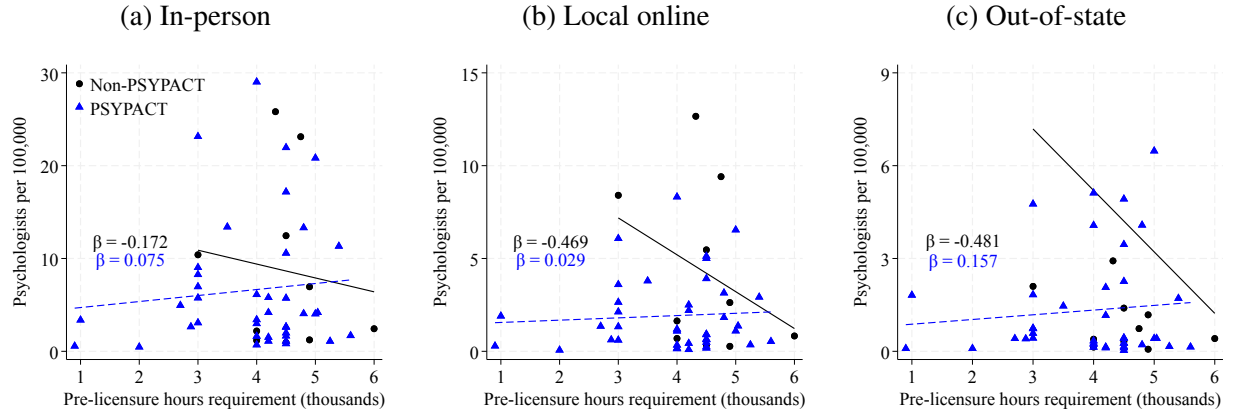
Notes: Data on PSYPACT membership come from the commission’s webpage: <https://psypact.gov/>

3 Descriptive evidence

Before turning to the structural model, I first present two pieces of descriptive evidence on how licensure requirements and licensure compacts affect entry. PSYPACT reduces the effective cost of interstate licensure by exempting member-state psychologists from satisfying the licensure require-

ments of each destination state. If this barrier is binding, then we would expect to see a negative relationship between licensure requirements and provider supply in non-PSYPACT states, particularly for out-of-state providers for whom such requirements are the sole barriers to entry. We would also expect to see this negative relationship attenuated in PSYPACT states as PSYPACT decouples out-of-state entry from the licensing stringency of the destination market. These predictions bear out in Figure 4, which provides correlations between provider supply and licensure requirements as measured by the number of supervised hours of practice that are required prior to licensure, separately by provider type and PSYPACT membership. In non-PSYPACT states, higher licensing hours are associated with lower equilibrium supply across all three provider types. In PSYPACT member states, these same slopes are near zero.

Figure 4: Correlation between psychologists per capita and pre-licensure hours requirement

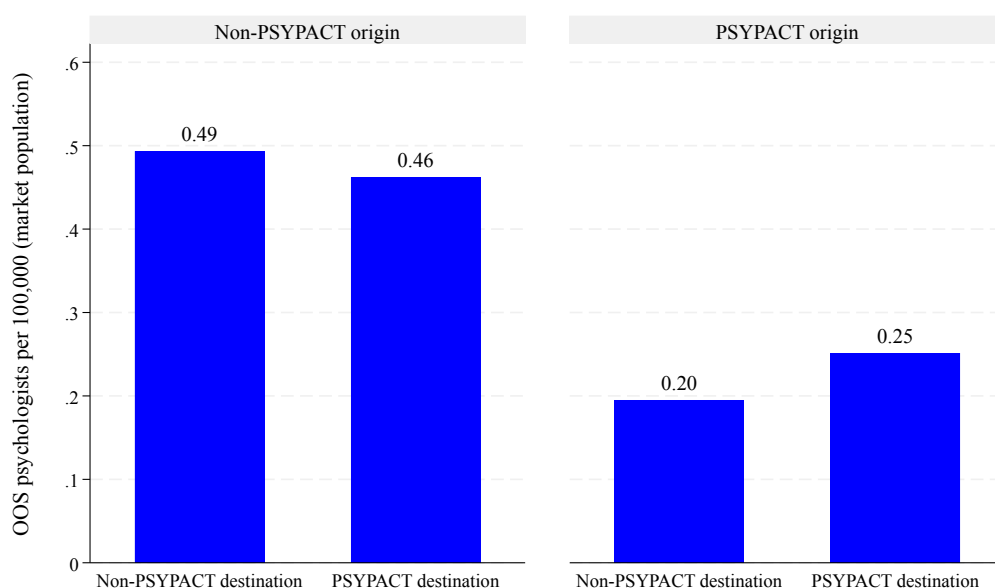


Notes: Each panel plots state-level psychologist supply per 100,000 residents against the state's pre-licensure hours requirement (in thousands), separately for PSYPACT member states (blue triangles) and non-member states (black circles). Fitted lines are estimated separately for each group via OLS. Reported slope coefficients β are normalized by the mean of non-PSYPACT states, so that units are interpretable as the percent change in provider supply per 1,000 licensing hours.

Second, PSYPACT reduces the cost of virtually entering other compact member states, conditional on residing in a compact member state. For this reason, we should expect to see that out-of-state providers from PSYPACT states are more likely to choose other PSYPACT member states as their destination than those from non-PSYPACT states. Figure 5 summarizes how out-of-state provider supply varies according to the PSYPACT membership status of the state from

which they originate. For each state pair, I compute the number of out-of-state providers per 100,000 destination-state residents and then average across pairs based on the PSYPACT membership status of the source and destination state. Among providers located in PSYPACT member states, entry is 25% higher in PSYPACT destinations than in non-PSYPACT destinations (+0.05 per 100,000). Among providers located in non-PSYPACT states, entry is 6% higher into non-PSYPACT destinations (+0.03 per 100,000). These differences suggest that PSYPACT impacts the flows of out-of-state providers, directing them to other member states.

Figure 5: Out-of-state psychologist supply by PSYPACT status of origin and destination state



Notes: Each bar reports the average number of out-of-state psychologist-market advertising relationships per 100,000 destination-state residents, averaged across state pairs within each cell.

These patterns together motivate a structural model of psychologist entry in which the profitability of entry is a function of PSYPACT membership, and the impact of compact membership is allowed to vary according to provider modality.

4 Entry model

I employ a static entry model of the simultaneous entry decisions of psychologists differentiated by modality and geographic proximity to the markets they serve in the style of [Bresnahan and Reiss \(1991\)](#). Let n_m^t denote the number of psychologists of type t in market m . All psychologists of a given type are identical and earn profits according to

$$\pi_m^t(\mathbf{N}_m, \mathbf{X}_m, \mathbf{W}_m^t) = S_m v^t(\mathbf{N}_m, \mathbf{X}_m) - F(n_m^t, \mathbf{W}_m^t).$$

Here, v^t corresponds to average variable profits, which are proportional to market population S_m and depend on the number of psychologists in the market $\mathbf{N}_m$ and market characteristics that determine demand and variable costs $\mathbf{X}_m$. Fixed costs F are a function of the number of firms of one's own type n_m^t as well as cost shifters $\mathbf{W}_m^t$. Licensure policies—including hours requirements, fees, and PSYPACT membership—are included as cost shifters in $\mathbf{W}_m^t$ for all psychologist types. Licensure requirements influence entry costs that do not scale with demand and so are fixed.

Entry thresholds are defined as the smallest market size that can sustain n^t providers of type t . This threshold is determined by the following break-even condition:

$$\pi_m^t(\mathbf{N}_m, \mathbf{X}_m, \mathbf{W}_m^t) = 0 \implies S_m^t \geq \frac{F(n_m^t, \mathbf{W}_m^t)}{v^t(\mathbf{N}_m, \mathbf{X}_m)}. \quad (1)$$

The first entry threshold S_1^t equals the ratio of fixed entry costs to variable profits per capita. Equation (1) illustrates that demand and cost shifters may result in entry below that which is indicated by clinical need alone. Factors that reduce fixed costs, including licensure compacts, lower S_1^t directly, expanding the set of markets in which entry is viable.

4.1 Identification and estimation

I empirically specify the profit function of a psychologist of type t as

$$\pi_m^t = S_m \left(\frac{\alpha^t}{n_m^t} + \beta^t \cdot \mathbf{1}[N_m^{-t} > 0] + \mathbf{X}_m \delta^t \right) - \mathbf{W}_m^t \theta^t - \sum_{n^t=1}^{N_t} \gamma_{n^t}^t + \varepsilon_m^t. \quad (2)$$

α^t and β^t capture how variable profits vary with the number of firms of one's own type and of other types, respectively. γ^t are components of fixed costs that depend on the number of firms of one's own type, and θ^t captures how fixed costs depend on the market's occupational licensure policy and other cost shifters. I parameterize own-type competition as $1/n_m^t$, so that $\alpha^t > 0$ implies variable profits are decreasing in the number of own-type competitors. Cross-type competition enters through an indicator for the presence of at least one competitor of another type because 53% of markets have no virtual providers of either type, leaving little variation beyond the extensive margin. $\beta^t < 0$ therefore implies that the presence of cross-type competitors reduces variable profits.

I impose the following assumptions on Equation (2):

- (i) Markets are independent. This implies that psychologists do not take into account the cross-market effects of entry, nor are there regional-level unobservables that correlate the payoffs of entry across markets, such as insurance networks. I follow the literature and minimize the scope for correlation in payoffs across markets by estimating my model using only geographically isolated markets. I follow [Magnolfi et al. \(2024\)](#) and define geographically isolated counties as those that are at least 35 miles away from all more populated counties. Table [A1](#) describes these 794 isolated counties. I test for robustness to these assumptions in Section 5.
- (ii) $\beta^t = 0$ for $t \in \{\text{out-of-state, local online}\}$. That is, out-of-state and local online psychologists have a competitive effect on in-person psychologists, but out-of-state and local online psychologists do not experience competitive pressure from psychologists of other types. This

assumption is motivated by a literature documenting the competitive effects of online entry on offline firms (Zervas et al., 2017; Seamans and Zhu, 2014; Goetz, 2023). Additionally, substitution to the marginal in-person psychologist will be heterogeneous across consumers within a market and will likely be confined to the set of consumers who are physically proximate to them, resulting in a relatively small effect of in-person entry on virtual profits. The same cannot be said for the marginal virtual psychologist, substitution to which does not decline in distance. This assumption is also made for modeling expediency: without it, the entry conditions across types are jointly determined and generally admit multiple equilibria.

- (iii) Unobserved profit shocks are normally distributed and are assumed to be equal for local online and in-person psychologists in each market. Additionally, the covariance of shocks for online and in-person psychologists with shocks to the profits of out-of-state psychologists is given by ρ :

$$(\varepsilon_m^{in} = \varepsilon_m^{onl}, \varepsilon_m^{out}) \sim \mathcal{N}(0, \Sigma)$$

$$\Sigma = \begin{bmatrix} 1 & 1 & \rho \\ 1 & 1 & \rho \\ \rho & \rho & 1 \end{bmatrix}$$

Profit shocks for in-person and local online providers are likely to be more tightly correlated with one another than with those faced by out-of-state psychologists because of their shared physical location: while shocks to demand are likely to generate correlation in the profits of all three types, local cost shocks will not affect the profitability of out-of-state psychologists, as their physical location is far removed from the market they have virtually entered.

Note that both ρ and β capture the correlation between n_m^{in} and n_m^{out} . Separate identification of these parameters requires the use of an instrument that shifts the profitability of one of these types of psychologists without directly influencing the profitability of the other. I use as a shifter of in-person psychologist profits a measure of each market’s cost-of-living, summarized in an “affordability index” that is computed as the ratio of price parities to median income. A higher

affordability index value indicates that the cost of living is high relative to what people earn and is a less affordable place to live. Because out-of-state providers do not live in the markets in which they practice, the costs of living in those markets are not costs that they themselves incur. Given that income itself is already included in the variable profit functions of all provider types, this measure is likely to satisfy the exclusion restriction and affect only in-person providers' profits. I model the affordability index as influencing in-person providers' fixed costs. Descriptive evidence of the instrument's strength can be seen in Table 1 in which the affordability index falls monotonically with the number of providers.

Estimates of θ_m^t , the effect of PSYPACT participation, are likely biased by correlation of PSYPACT participation with ε_m^t : e.g., participation in PSYPACT may be just one of many ways that a market endeavors to lessen constraints on psychologist supply, the rest of which are unobservable. In order to be able to compute unbiased estimates of the equilibrium effects of licensure compacts, I must address the potential endogeneity from correlated entry and PSYPACT participation choices. I do so by using the number of interstate compacts that each state participated in prior to PSYPACT as an instrument for PSYPACT participation. More pre-PSYPACT compact participation indicates that a state is more receptive to licensure reform. None of the pre-PSYPACT compacts pertained to mental health professions, and so participation in them is unlikely to affect psychologist supply except for its association with a state's propensity to participate in interstate compacts in general. Figure A2 illustrates the number of pre-PSYPACT compacts that each state participated in, which ranges from zero to four.

Because the probability that a market has a given number of entrants is nonlinear in the endogenous variable, the standard instrumental variables approach of substituting predicted values of the endogenous regressor into the profit equations would induce bias (Terza et al., 2008). Instead, I employ a control function approach, in which the residuals from the first-stage regression are used to approximate the component of ε_m^t that is correlated with PSYPACT participation.² A

²Use of the control function approach to address discrete endogenous explanatory variables requires that the first-stage residuals "act as a kind of sufficient statistic for capturing the endogeneity of [the discrete endogenous explanatory variable]" (Wooldridge, 2010). However, Wooldridge (2010) also points out that this assumption is "no more or less general than the bivariate probit assumption" made in the alternative maximum likelihood approach.

flexible function of these residuals is then included in the linear predictor of the second-stage ordered probit model (Petrin and Train, 2010; Wooldridge, 2010). In practice, I regress an indicator for PSYPACT participation on the components of the variable profit functions and fixed effects for the number of firms of each type. I then include a third-degree polynomial expansion of the residuals from each of these regressions in the profit functions for their respective types. Table A2 provides the main results of this first stage regression. The joint F -statistic for the number of pre-PSYPACT compact indicators is 79.7.

I estimate the model by maximum likelihood. When doing so, I restrict the set of values that n^t can take to $n^t = 0, 1, \dots, 4, 5$ to account for outliers.

5 Results

Table 2 provides parameter estimates for the entry model. Columns (1)–(3) provide estimates from univariate models in which the correlation of unobserved profit shocks across provider types is assumed to be zero. Columns (4)–(6) provide estimates from the bivariate model described in Section 4. Ordered probit coefficients, standard errors, and marginal effects are reported for each covariate. Marginal effects represent the partial equilibrium percent change in the number of providers associated with a one standard deviation increase in the covariate.

In Table 2, we see that the estimated correlation between unobservables equals 0.55, indicating that unobserved profit shocks are tightly correlated across provider types. Given that the model is nonlinear, there is no formal diagnostic test for instrument strength. I note that the first-stage F -statistic on the number of pre-PSYPACT compacts in Table A2 is 79.7, providing confidence in the relevance of the instrument. The affordability index, which serves as the exclusion restriction identifying the in-person equation separately from the correlation in unobservables, has a large and statistically significant effect on in-person provider supply, a one standard deviation increase resulting in a 4.7% decline in the number of in-person psychologists. In the univariate estimates in Column (1), we see that the presence of a virtual provider lowers the number of in-person providers

by a modest 1.4%. This estimate is attenuated relative to that from Column (4), which accounts for the endogeneity of virtual entry and indicates that the presence of a virtual provider lowers in-person supply by 18.4%.

Across all provider types, PSYPACT increases entry. Effects are particularly large for out-of-state providers, for which a market's participation in PSYPACT increases entry by 27.5%. That the direct effect of PSYPACT on in-person providers is positive suggests that many of these out-of-state entrants also offer in-person services in their home state, and that substitution away from in-person provision is outweighed by the entry of new psychologists into the market who, having found virtual provision sufficiently profitable, also choose to offer in-person services.

Included as measures of fixed costs alongside PSYPACT membership are two measures of licensure requirements: the number of pre-licensure supervised hours and licensure fees. That these estimates are small in magnitude, imprecise, and in many cases positive is reflective of the importance of addressing the endogeneity of licensure policies.

Table [A3](#) presents the remaining estimates for the socio-demographic variable profit parameters and fixed cost parameters. Several notable differences emerge across outcomes in how demographics affect entry. In-person and local online supply are both positively correlated with the share of insured individuals, perhaps because out-of-state providers are effectively out-of-network for patients in the destination market, making local insurance coverage rates less relevant to their entry decisions. By contrast, only in-person provider supply is lower in markets with higher shares of Black individuals; no such relationship emerges for out-of-state or local online providers. Provider supply across all three types is lower in markets with higher shares of adults without college degrees, though the effect is considerably smaller for out-of-state providers. As with the licensure requirement estimates, I note that these variable profit parameter estimates are purely correlational.

Table 2: Entry model estimates

	Univariate model			Bivariate model		
	In-person	Out-of-state	Local online	In-person	Out-of-state	Local online
	(1)	(2)	(3)	(4)	(5)	(6)
Correlation in profit shocks (ρ)	0.550 (0.161)					
Variable profit parameters						
Number of own-type competitors (α)	0.100 (0.079) —	0.002 (0.011) —	0.005 (0.029) —	0.111 (0.074) —	0.014 (0.012) —	0.054 (0.028) —
Presence of other-type competitor(s) (β^{in})	−0.004 (0.044) −0.014				−0.092 (0.030) −0.184	
Fixed cost parameters						
Affordability index (θ^{in})	−0.455 (0.130) −0.070				−0.497 (0.137) −0.047	
PSYPACT (θ_t)	1.286 (0.378) 0.198	0.934 (0.428) 0.650	0.734 (0.435) 0.165	1.733 (0.462) 0.163	0.820 (0.405) 0.275	0.163 (0.413) 0.031
Hours (θ_t)	0.614 (0.161) 0.095	0.303 (0.157) 0.211	0.389 (0.166) 0.088	0.731 (0.169) 0.069	0.227 (0.111) 0.076	0.249 (0.152) 0.047
Fee (θ_t)	−0.096 (0.121) −0.015	0.055 (0.099) 0.038	−0.119 (0.123) −0.027	−0.043 (0.139) −0.004	0.174 (0.094) 0.059	−0.053 (0.100) −0.010
Log likelihood	−516.07	−551.85	−502.43	−1,569.28		
Number of observations	793	793	793	793		

Notes: Columns (1)–(3) present results for univariate models, while Columns (4)–(6) present estimates for the bivariate model. Results are presented on two lines, the first line providing the coefficient and its standard error in parentheses and the second line providing the corresponding marginal effect. Standard errors are calculated using 100 bootstrap replications, while marginal effects are the simulated percent change in the number of psychologists given a one standard deviation in the row variable.

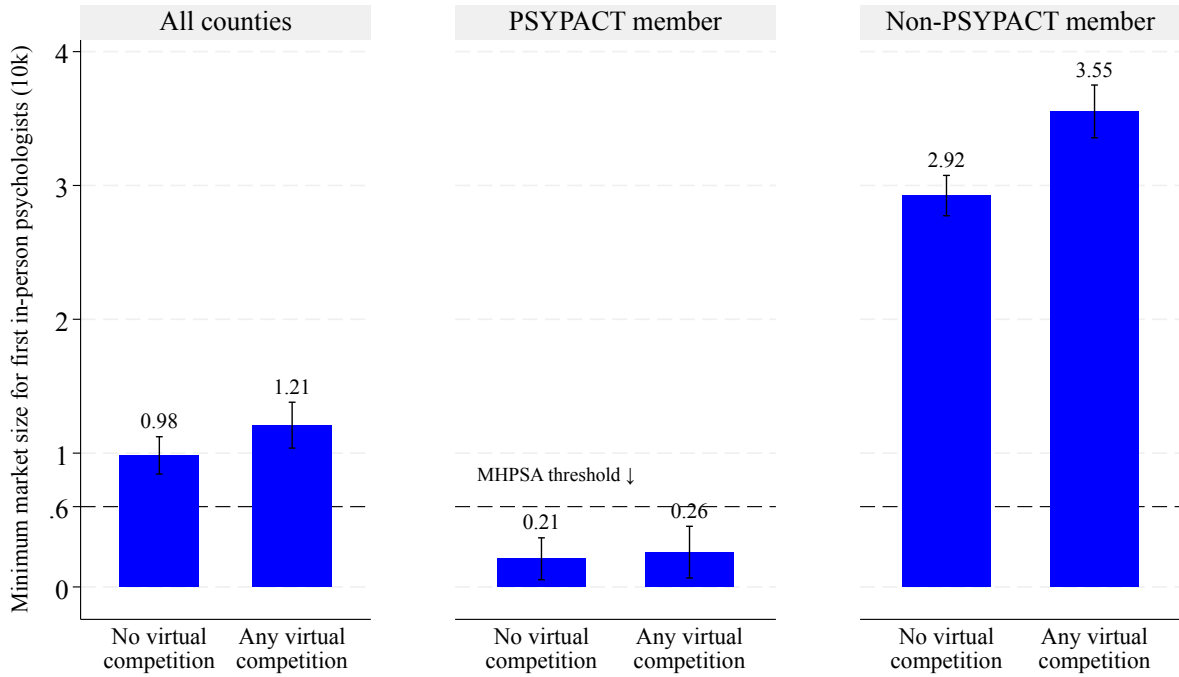
5.1 Entry thresholds and access to care

Using the model estimates, I compute entry thresholds for the first provider according to Equation (1). These thresholds can be interpreted as the minimum market size, in ten thousand individuals per firm, necessary to sustain at least one provider. Figure 6 reports S_1 for in-person psychologists, stratified by PSYPACT membership status and the presence of virtual competition. I compare this threshold to that used for Mental Health Professional Shortage Area (MHPSA) designations. Counties are designated as MHPSAs if they fall below a population-to-provider ratio of 6,000 (U.S. Department of Health and Human Services, 2024). Psychiatrists, psychologists, clinical social workers, psychiatric nurse specialists, and marriage and family therapists are included in the count of providers used to construct this threshold. While the MHPSA ratio counts all five provider types, Table A1 shows that 77.7% of isolated counties with no in-person psychologist carry an MHPSA designation, confirming that psychologist supply is strongly predictive of shortage status across the full provider count.

The leftmost panel provides entry thresholds evaluated at the mean rate of PSYPACT participation across all counties in the sample. In the absence of virtual competition, S_1 is approximately 9,800 individuals, which is just above the MHPSA designation cutoff of 6,000 individuals per provider. This proximity to the policy threshold rationalizes the pattern of MHPSA designations in the data: variation in local demand conditions and licensing environments places some counties above and others below the cutoff.

The remaining panels reveal that PSYPACT membership substantially shifts this threshold. In PSYPACT-member states, the minimum market size for the first in-person psychologist is only approximately 2,100 individuals, which is well below the MHPSA cutoff and implies that nearly any market can sustain at least one provider. In non-member states, the corresponding threshold rises to approximately 29,200 individuals, nearly five times the MHPSA cutoff, consistent with substantially higher fixed costs of entry in the absence of an interstate licensure compact. The presence of virtual competition raises thresholds in both groups, as virtual providers reduce the

Figure 6: Entry thresholds for first in-person psychologist



Notes: Figure reports the minimum market size necessary to result in the entry of the first in-person psychologist. Thresholds are measured in tens of thousands of individuals. The dashed line indicates the Mental Health Professional Shortage Area (MHPSA) designation threshold of one provider per 6,000 individuals. Error bars denote 95% confidence intervals from 100 bootstrap replications.

variable profits available to potential in-person entrants. Taken together, these results suggest that interstate licensure compacts are a particularly powerful lever for expanding access in isolated markets: PSYPACT membership alone is sufficient to bring the minimum viable market size well below the threshold for shortage designation.

5.2 Robustness

I check the robustness of these results to the definition of geographically isolated markets. My main analysis uses counties at least 35 miles from any more populous county. I construct an alternative analysis sample using a more conservative criterion of 40 miles. This stricter threshold yields 626 counties, a 21% decrease in sample size relative to the main analysis. Table A4 presents model estimates for the 40-mile subsample. Estimates for competition effects, PSYPACT, and the

correlation in profit shocks are qualitatively similar relative to the main results.

I also conduct a placebo test to evaluate my product market definition. I estimate an otherwise identical model replacing the count of in-person psychologists with the total count of licensed marriage and family therapists (LMFTs) per county. LMFTs specialize in relational and family therapy and cannot perform psychological assessment. PSYPACT applies specifically to psychologists and has no scope to directly affect LMFT entry. If LMFTs are also poor substitutes for psychologists, then there is no indirect competitive channel either, leaving no channel for PSYPACT to affect LMFT supply. Results are presented in Table A5. The estimated marginal effect of PSYPACT on the expected number of LMFTs is 0.010, which is an order of magnitude smaller than the corresponding estimate of 0.198 for psychologists and not statistically significant. This supports the interpretation that the main result is driven by the specific mechanism of the PSYPACT compact as well as my product market definition.

6 Equilibrium effects of occupational licensure reform

The results above suggest that there is scope for interstate licensure compacts like PSYPACT to both reduce entry costs and increase the competitive pressure placed on in-person psychologists from out-of-state practitioners. In this section, I decompose the total effect of PSYPACT on in-person psychologist supply into what I refer to as direct and indirect effects. The direct effect refers to PSYPACT's effect on profits, holding fixed counts of competitors of other types. For in-person providers, the direct effect captures how PSYPACT increases the profitability of in-person entry by enabling providers to supplement local practice with telehealth services delivered to patients in other member states. The indirect effect refers to increased competitive pressure resulting from the entry of virtual competitors. The total effect of PSYPACT is the sum of the two.

I compute the effect of implementing PSYPACT for the subsample of markets where it has not been implemented. I further restrict this subsample to markets with no more than 10 in-person providers in order to exclude observations in the tails of the distribution of in-person provider

counts. I proceed in the following steps:

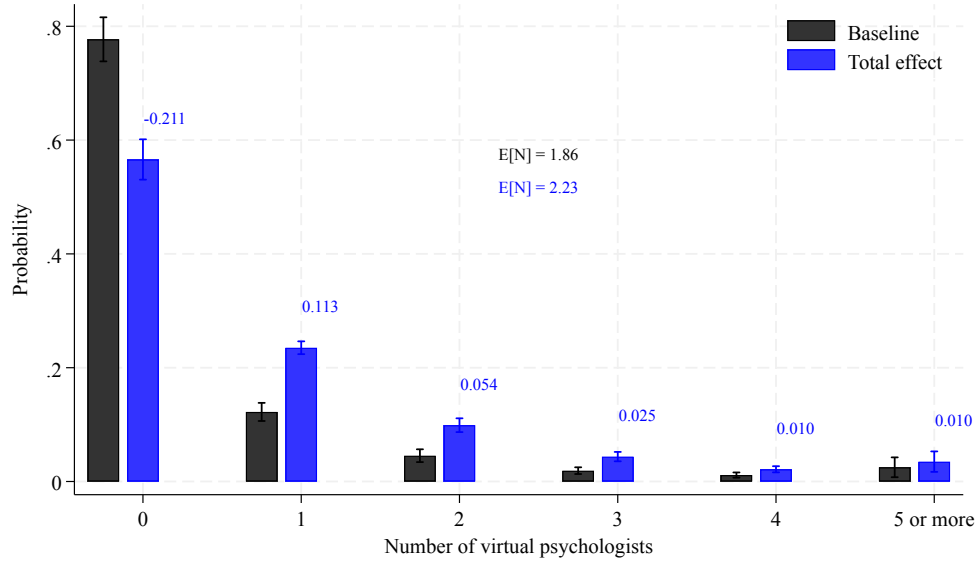
- (i) Compute the equilibrium effect of PSYPACT on out-of-state and local online entry by setting the PSYPACT indicator to one in each type's profit function, predicting the number of providers of each type, substituting this prediction in for n_m^t , and iterating until convergence at n^{*out} and n^{*onl} .
- (ii) Compute the direct equilibrium effect of PSYPACT on in-person entry by setting the PSYPACT indicator to one, holding competitor counts at their observed values, predicting the number of in-person providers, substituting this prediction in for n^{in} , and iterating until convergence at $n^{*direct}$.
- (iii) Compute the total equilibrium effect of PSYPACT on in-person entry by setting the PSYPACT indicator to one and $N^{-t} = n^{*out} + n^{*onl}$, predicting the number of in-person providers, substituting this prediction in for n^{in} , and iterating until convergence at n^{*total} .

Figure 7 summarizes the results of this routine, showing the probability that the number of psychologists of a given type equals a given value. Panel (a) shows results for virtual psychologists (those practicing out-of-state or locally online), with black bars providing the model's predicted likelihood of observing each level of supply for the baseline sample in which PSYPACT is not active. Blue bar labels correspond to the difference between the counterfactual probability and the probability at baseline. The activation of PSYPACT substantially shifts probability mass away from zero: the probability of a market having no virtual psychologists falls from 77.7% to 56.6%, and the expected number of virtual psychologists rises by 0.37 (20%).

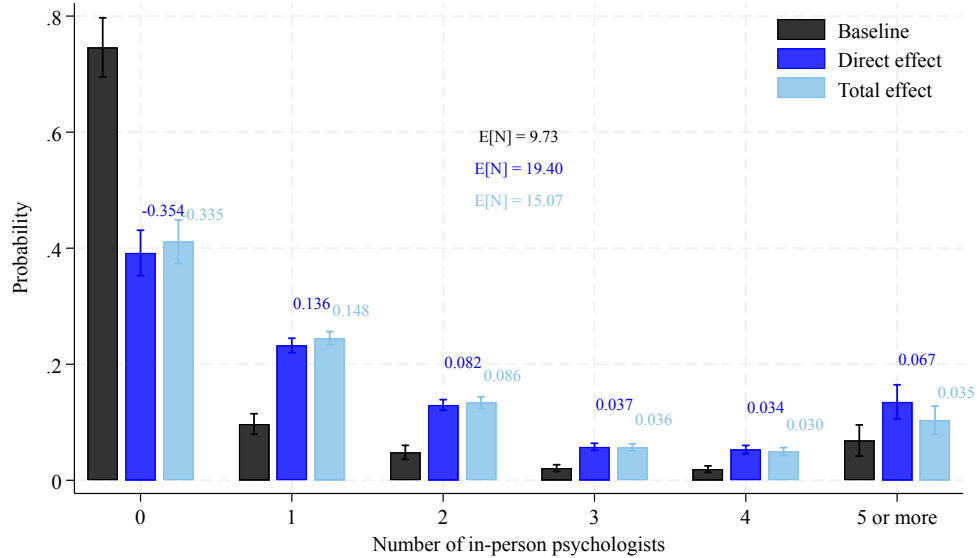
Panel (b) shows results for in-person psychologists. The direct effect of PSYPACT is to substantially increase in-person supply: the probability of there being no in-person psychologists in a market falls by 35.4 p.p. and the expected number of in-person psychologists nearly doubles, rising by 9.67 on average. The total effect, which takes into account the indirect competitive effect of virtual entry, is attenuated: the estimated total effect of PSYPACT on in-person psychologist supply is an increase of 5.34 psychologists (55%) on average. The indirect effect of virtual entry

Figure 7: Decomposition of PSYPACT's effect on psychologist supply

(a) Virtual (out-of-state + local online)



(b) In-person



Notes: Figure illustrates counterfactual effect of implementing PSYPACT in counties where it is not already in place. Bars provide the predicted probability of a market having the number of psychologists indicated on the x-axis. $E[N]$ denotes the expected number of psychologists of the given type per market. Baseline predicted probabilities are those at observed values of covariates. Direct effect refers to predicted probabilities in equilibrium when PSYPACT is activated. Total effect refers to predicted probabilities in equilibrium when PSYPACT is activated and competitor counts are set to their predicted value in equilibrium. Standard errors calculated using 100 bootstrap replications.

amounts to a reduction of 4.33 providers—45% of the direct gain—reflecting the competitive pressure exerted by the additional virtual entrants. Note that without taking into account the modality or geographic locations of providers, PSYPACT is unambiguously successful in increasing the supply of psychotherapists, resulting in an average net increase of 5.71 providers per market.

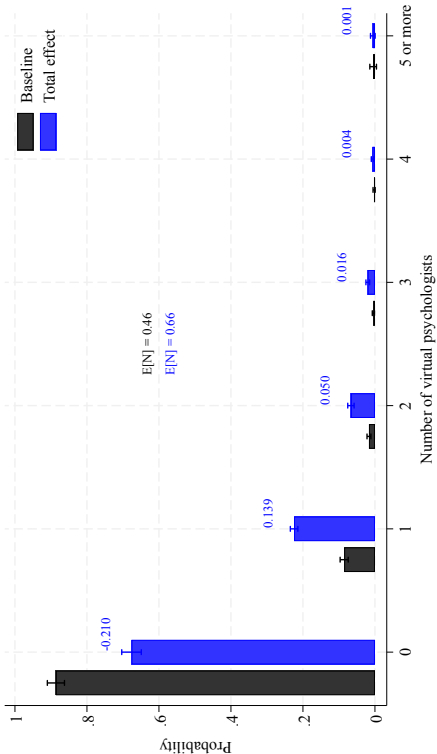
The indirect effect of virtual entry on in-person provider supply has interesting implications not just for the average effect of PSYPACT, but also for the distribution of effects across markets. In particular, these indirect effects are likely to be most salient in markets that already had a non-zero number of in-person psychologists *ex ante*, so that there is scope for virtual entry to induce in-person exit. Figure 8 explores this heterogeneity by stratifying markets based on whether their observed number of in-person psychologists equals zero. The baseline predicted difference in total psychologist counts between markets with at least one in-person provider and those with none is 34.70, of which 86% (29.88 providers) is attributable to differences in in-person supply.

For in-person providers, the direct effect of PSYPACT substantially increases supply in both groups. In markets with no baseline in-person psychologists, the direct effect raises the predicted number of in-person providers by 6.22 (from 1.01 to 7.23). In markets with at least one in-person provider, the predicted number of in-person providers rises from 30.89 to 48.92 (58%). The indirect competitive effect, however, operates very differently across groups. Among markets with zero baseline in-person providers, virtual entry reduces the predicted in-person count by only 1.27 providers (20% of the direct gain). For markets with at least one in-person provider, the indirect effect amounts to a reduction of 11.73 providers (65% of the direct gain). The total effect of PSYPACT on in-person supply is accordingly an increase of 4.95 providers (490%) in the zero-baseline group, compared to 6.30 providers (20%) in the at-least-one group.

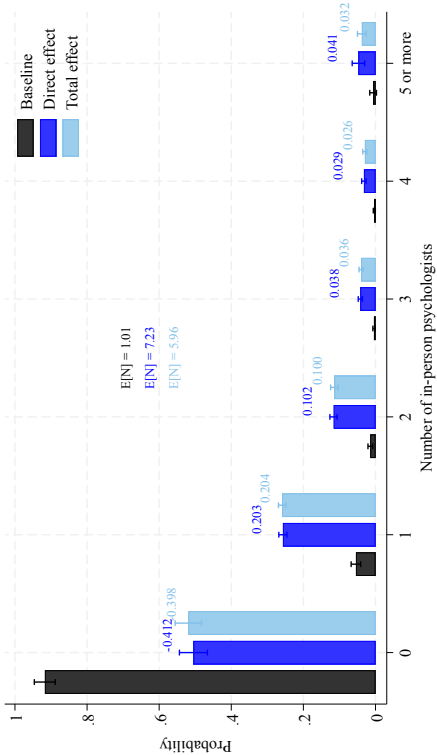
Figure 8: Decomposition of PSYPACT's effect on psychologist supply, by baseline in-person provider presence

Panel (a): Zero baseline in-person providers

(a1) Virtual (out-of-state + local online)

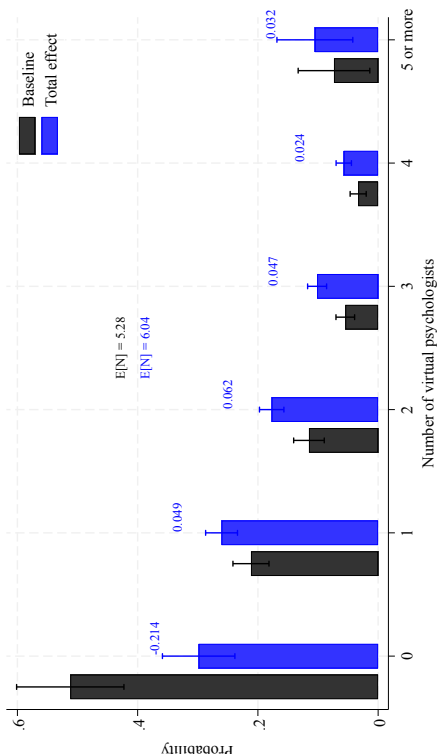


(a2) In-person

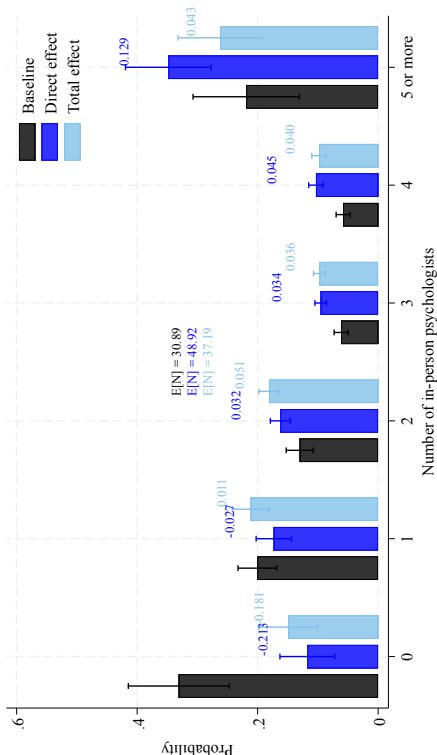


Panel (b): ≥ 1 baseline in-person provider

(b1) Virtual (out-of-state + local online)



(b2) In-person



Notes: Figure illustrates counterfactual effect of implementing PSYPACT in counties where it is not already in place, stratified by whether the market has zero or at least one observed in-person psychologist. Bars provide the predicted probability of a market having the number of psychologists indicated on the x-axis. $E[N]$ denotes the expected number of psychologists of the given type per market. Baseline predicted probabilities are those at observed values of covariates. Direct effect refers to predicted probabilities in equilibrium when PSYPACT is activated. Total effect refers to predicted probabilities in equilibrium when PSYPACT is activated and competitor counts are set to their predicted value in equilibrium. Standard errors calculated using 100 bootstrap replications.

The concentration of indirect crowding-out in already high-supply markets has important implications for the gap between the two groups. The direct effect of PSYPACT alone would widen the in-person provider gap from 29.88 to 41.69. The indirect competitive response, however, substantially attenuates this widening: after accounting for the competitive effects of virtual entry, the gap narrows from 41.69 to 31.23—only marginally above the baseline gap of 29.88. Crowding-out thus occurs most heavily where it is least consequential: in markets already well-supplied with in-person providers.

This finding that PSYPACT is most beneficial for markets that would ex-ante be classified as most in need of treatment is neither mechanical nor the result of bias from endogenous stratification ([Abadie et al., 2018](#)). Instead, it demonstrates that whether and how reductions in entry barriers affect provider supply depends on the presence and intensity of competition from differentiated competitors.

7 Conclusion

Using a novel dataset on mental health provider supply, this paper demonstrates that interstate licensing compacts have the potential to affect the size and composition of mental healthcare markets. PSYPACT successfully increases overall psychologist supply, expanding access to both virtual and in-person care. Indirect competitive pressure exerted by virtual entrants dampens in-person supply gains, offsetting 45% of the direct effect, but this crowding-out falls disproportionately on markets that were already well-supplied. As a result, PSYPACT generates its largest proportional gains in in-person supply precisely in markets with the greatest pre-existing shortages. As policymakers continue to pursue licensing compacts as a solution to healthcare workforce shortages, these results suggest that the distributional consequences of such reforms depend critically on the pre-existing competitive landscape, and that equilibrium responses among differentiated provider types can meaningfully reshape where the benefits of reform accrue. This is particularly important given the impending implementation of the Counseling Compact, an interstate licensure

compact for Licensed Professional Counselors, which is currently being implemented ([Counseling Compact, 2025](#)). Connecting these supply-side changes to patient utilization and mental health outcomes remains an important direction for assessing the full welfare consequences of licensure reform.

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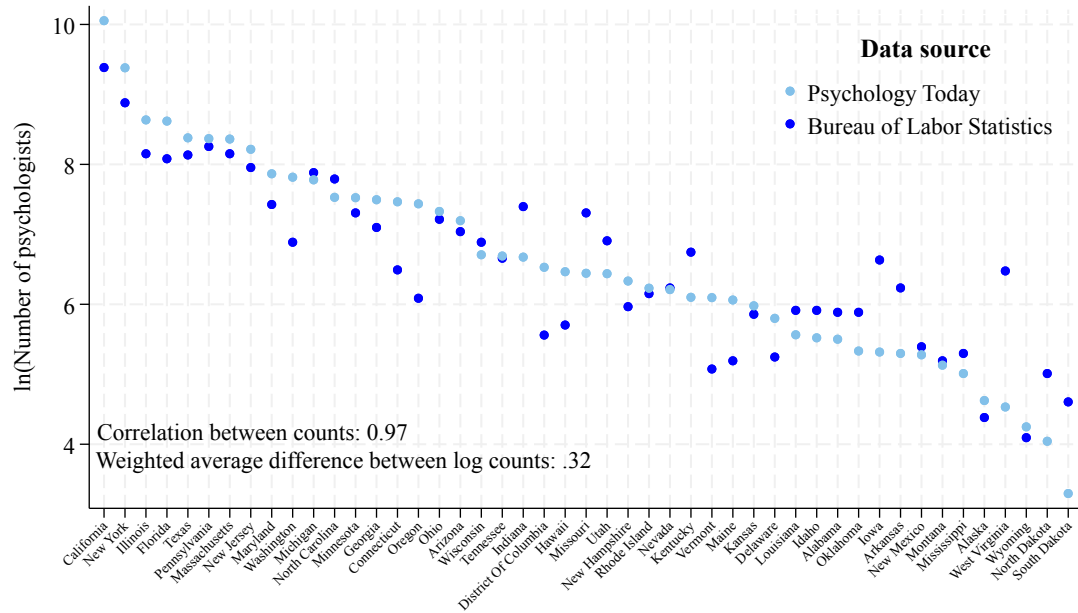
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Appendix

Figure A1: Comparison of psychologist counts across data sources



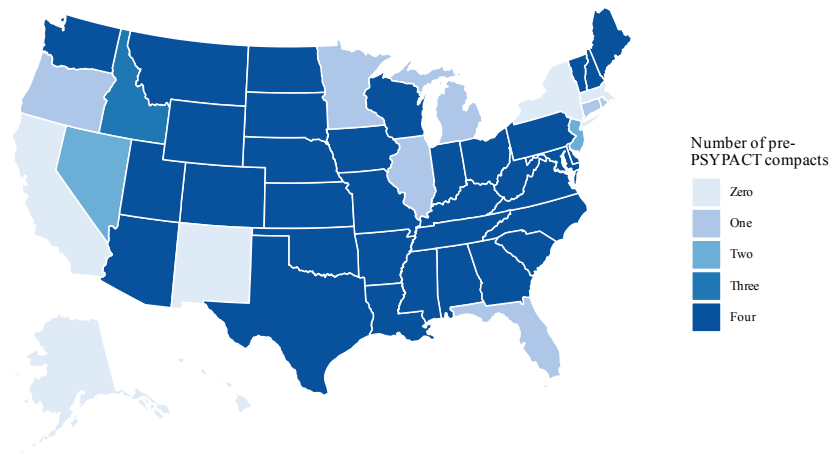
Notes: Data come from the Bureau of Labor Statistics Occupational Employment and Wage Statistics: <https://data.bls.gov/oes>. Difference in log counts is weighted by state population.

Table A1: Isolated county characteristics by number of in-person psychologists

	Number of in-person psychologists					
	0	1	2	3	4	≥ 5
Competitor counts						
Out-of-state	0.1 (0.3)	0.2 (0.5)	0.3 (1.0)	0.7 (1.0)	0.7 (1.2)	11.1 (23.8)
Local online	0.1 (0.4)	0.5 (1.0)	0.4 (0.8)	1.2 (1.5)	1.8 (2.0)	51.1 (160)
Policy environment						
PSYPACT	71.1%	75.7%	72.5%	78.9%	81.0%	74.4%
Pre-PSYPACT compact count	3.1 (1.5)	3.0 (1.6)	3.0 (1.6)	2.8 (1.8)	3.1 (1.6)	2.6 (1.7)
Hours (thous.)	3.9 (1.2)	4.0 (1.3)	4.2 (0.9)	3.9 (0.8)	4.0 (1.2)	4.1 (0.9)
Fees (\$h)	2.4 (1.0)	2.3 (1.0)	2.5 (1.1)	2.7 (1.4)	2.4 (1.3)	2.4 (1.2)
MHPSA designation rate	77.7%	67.1%	65.0%	68.4%	42.9%	33.2%
Socio-demographics						
Population (10k)	2.5 (3.0)	5.2 (4.1)	7.1 (5.1)	9.9 (10.5)	11.7 (7.0)	59.6 (94.7)
Median income (\$10k)	6.2 (1.2)	6.5 (1.2)	6.4 (1.2)	7.3 (1.3)	6.5 (0.8)	7.7 (1.7)
Share insured	89.1 (6.1)	90.3 (4.2)	90.3 (5.1)	90.8 (3.9)	90.7 (4.7)	91.6 (4.2)
Share unemployed	4.7 (3.4)	4.7 (2.3)	4.9 (2.2)	4.5 (1.5)	5.2 (1.9)	4.8 (1.5)
Share Black	4.3 (10.2)	6.0 (11.5)	7.3 (13.3)	5.8 (11.4)	12.5 (15.4)	10.5 (12.0)
Share Hispanic	15.3 (20.3)	15.0 (16.8)	12.4 (19.2)	13.0 (14.5)	10.6 (6.9)	17.2 (17.7)
Share over 65	20.7 (6.3)	19.3 (5.2)	19.5 (4.4)	18.9 (5.0)	17.8 (4.0)	17.3 (4.5)
Share $\leq$ high school	33.3 (5.8)	31.6 (5.1)	30.7 (3.8)	29.1 (7.3)	28.5 (5.4)	24.7 (5.2)
Share rural	45.5 (41.8)	22.9 (27.2)	18.7 (27.4)	13.0 (21.3)	9.5 (20.9)	5.8 (13.0)
Share with broadband	83.1 (6.4)	85.8 (4.6)	84.9 (6.6)	88.4 (3.5)	87.7 (2.5)	90.0 (2.6)
Affordability index	100.2 (0.9)	100.0 (0.8)	100.1 (0.8)	99.5 (0.5)	100.0 (0.6)	99.4 (0.6)
Observations	394	70	40	19	21	250
Share of all observations	49.6	8.8	5.0	2.4	2.6	31.5

Notes: Isolated counties are those with population-weighted centroids that are more than 35 miles away from all more populated counties. In-person psychologist and competitor counts come from Psychology Today. PSYPACT is a binary indicator for participation in the Psychology Interjurisdictional Compact. The affordability index is computed using price parity and income data from the Bureau of Economic Analysis (BEA, 2024). Pre-PSYPACT interstate licensure compacts include the Nurse Licensure Compact (2000), EMS Compact (2017), Physical Therapy Compact (2017), and Interstate Medical Licensure Compact (2017). All socio-demographic characteristics come from the American Community Survey (Census Bureau, 2025). MHPSA stands for Mental Health Professional Shortage Area.

Figure A2: Pre-PSYPACT interstate licensure compact participation



Notes: Pre-PSYPACT interstate licensure compacts include the Nurse Licensure Compact (2000), EMS Compact (2017), Physical Therapy Compact (2017), and Interstate Medical Licensure Compact (2017).

Table A2: First stage estimates

PSYPACT		
<i>Number of pre-PSYPACT compacts</i>		
Zero	(ref)	
One	0.723**	(0.217)
Two	1.073***	(0.072)
Three	0.854***	(0.102)
Four	0.881***	(0.087)
<i>F</i> -statistic	79.68	
Observations	793	

Notes: Regression includes county demographic and economic characteristics, state licensure requirements, market competition counts, and provider count indicators. Standard errors clustered at the state-level in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A3: Entry model estimates continued

	Univariate model			Bivariate model		
	In-person (1)	Out-of-state (2)	Local online (3)	In-person (4)	Out-of-state (5)	Local online (6)
Variable profit parameters						
Median income (δ)	-0.034 (0.027)	-0.012 (0.003)	-0.003 (0.014)	-0.046 (0.032)	-0.017 (0.005)	0.002 (0.021)
Share insured (δ)	0.025 (0.013)	0.002 (0.005)	0.020 (0.008)	0.038 (0.016)	0.003 (0.008)	0.026 (0.012)
Share unemployed (δ)	0.015 (0.023)	-0.009 (0.008)	-0.009 (0.014)	0.013 (0.028)	-0.016 (0.010)	-0.023 (0.019)
Share Black (δ)	-0.022 (0.009)	-0.002 (0.004)	-0.004 (0.006)	-0.028 (0.009)	-0.003 (0.006)	-0.002 (0.007)
Share Hispanic (δ)	-0.021 (0.009)	-0.007 (0.005)	-0.013 (0.007)	-0.022 (0.012)	-0.011 (0.006)	-0.016 (0.009)
Share over 65 (δ)	0.005 (0.017)	0.012 (0.005)	0.007 (0.010)	0.009 (0.018)	0.015 (0.007)	0.013 (0.011)
Share $\leq$ high school (δ)	-0.078 (0.012)	-0.024 (0.005)	-0.062 (0.009)	-0.106 (0.015)	-0.031 (0.007)	-0.077 (0.011)
Share rural (δ)	0.050 (0.031)	0.039 (0.027)	0.088 (0.028)	0.070 (0.039)	0.037 (0.029)	0.104 (0.028)
Share with broadband (δ)	0.012 (0.013)	0.010 (0.005)	0.004 (0.010)	0.018 (0.014)	0.014 (0.008)	0.002 (0.012)
Fixed cost parameters						
γ_1^t	-0.171 (0.653)	1.875 (0.288)	1.756 (0.346)	0.465 (0.682)	2.614 (0.444)	2.261 (0.492)
γ_2^t	0.460 (0.655)	2.584 (0.302)	2.382 (0.372)	1.297 (0.695)	3.600 (0.453)	3.120 (0.503)
γ_3^t	0.877 (0.657)	3.064 (0.314)	2.865 (0.403)	1.859 (0.699)	4.256 (0.452)	3.840 (0.536)
γ_4^t	1.101 (0.661)	3.443 (0.322)	3.182 (0.412)	2.166 (0.702)	4.774 (0.469)	4.307 (0.555)
γ_5^t	1.347 (0.653)	3.666 (0.329)	3.511 (0.408)	2.508 (0.719)	5.087 (0.478)	4.746 (0.535)
Log likelihood	-516.07	-551.85	-502.43		-1,569.28	
Number of observations	793	793	793		793	

Notes: Columns (1)-(3) present results for univariate models, while Columns (4)-(6) present estimates for the bivariate model. Each cell contains the ordered probit coefficient and its standard error in parentheses. Standard errors are calculated using 100 bootstrap replications.

Table A4: Entry model estimates using more conservative definition of isolated counties

	Univariate model			Bivariate model		
	In-person (1)	Out-of-state (2)	Local online (3)	In-person (4)	Out-of-state (5)	Local online (6)
Correlation between error terms (ρ)	0.575 (0.137)					
Variable profit parameters						
Own-type competition (α^t)	0.079 (0.071)	0.007 (0.014)	0.007 (0.037)	0.090 (0.063)	0.017 (0.016)	0.041 (0.028)
Other-type competition (β^{in})	0.014 (0.031)			-0.066 (0.032)		
Median income (δ)	-0.041 (0.023)	-0.008 (0.005)	-0.001 (0.017)	-0.052 (0.028)	-0.010 (0.006)	0.001 (0.021)
Share insured (δ)	0.022 (0.015)	0.001 (0.006)	0.021 (0.008)	0.034 (0.016)	0.001 (0.008)	0.027 (0.012)
Share unemployed (δ)	0.013 (0.028)	-0.009 (0.010)	-0.014 (0.016)	0.011 (0.028)	-0.015 (0.012)	-0.025 (0.021)
Share Black (δ)	-0.022 (0.009)	-0.000 (0.004)	-0.002 (0.008)	-0.027 (0.012)	-0.001 (0.005)	-0.002 (0.008)
Share Hispanic (δ)	-0.024 (0.014)	-0.007 (0.005)	-0.011 (0.008)	-0.025 (0.013)	-0.011 (0.006)	-0.015 (0.010)
Share over 65 (δ)	-0.001 (0.020)	0.014 (0.005)	0.004 (0.011)	0.004 (0.022)	0.018 (0.008)	0.008 (0.012)
Share $\leq$ high school (δ)	-0.074 (0.013)	-0.022 (0.006)	-0.060 (0.010)	-0.100 (0.019)	-0.029 (0.007)	-0.071 (0.013)
Share rural (δ)	0.042 (0.027)	0.030 (0.027)	0.086 (0.026)	0.060 (0.033)	0.032 (0.026)	0.111 (0.035)
Share with broadband (δ)	0.018 (0.015)	0.009 (0.006)	0.003 (0.011)	0.023 (0.018)	0.013 (0.008)	0.002 (0.013)
Fixed cost parameters						
Affordability index (θ^{in})	-0.373 (0.112)			-0.412 (0.125)		
PSYPACT (θ_t)	0.975 (0.364)	0.830 (0.516)	0.608 (0.454)	1.469 (0.451)	0.869 (0.545)	0.331 (0.419)
Hours (θ_t)	0.407 (0.134)	0.235 (0.181)	0.257 (0.168)	0.528 (0.143)	0.218 (0.136)	0.178 (0.125)
Fee (θ_t)	-0.053 (0.122)	0.023 (0.116)	-0.053 (0.124)	-0.013 (0.156)	0.093 (0.114)	0.004 (0.100)
γ_1^t	-0.168 (0.596)	1.629 (0.342)	1.579 (0.385)	0.413 (0.665)	2.322 (0.471)	2.099 (0.480)
γ_2^t	0.384 (0.600)	2.357 (0.371)	2.236 (0.415)	1.176 (0.679)	3.348 (0.488)	3.038 (0.468)
γ_3^t	0.716 (0.621)	2.817 (0.393)	2.730 (0.430)	1.642 (0.706)	3.983 (0.496)	3.777 (0.503)
γ_4^t	0.914 (0.635)	3.219 (0.423)	3.065 (0.456)	1.919 (0.709)	4.535 (0.517)	4.264 (0.514)
γ_5^t	1.177 (0.647)	3.431 (0.418)	3.435 (0.476)	2.289 (0.722)	4.828 (0.506)	4.765 (0.543)
Log likelihood	-436.971	-458.790	-439.957		-1311.109	
Number of observations	626	626	626		626	

Notes: Analysis sample includes counties at least 40 miles from the all more populous counties. Columns (1)-(3) present results for univariate models, while Columns (4)-(6) present estimates for the bivariate model. Results are presented on two lines, the first line providing the coefficient and its standard error in parentheses and the second line providing the corresponding marginal effect. Standard errors are calculated using 100 bootstrap replications, while marginal effects are the simulated percent change in the number of psychologists given a one standard deviation in the row variable.

Table A5: Placebo test estimates

	All LMFTs
Variable profit parameters	
Number of own-type competitors (α)	0.009 (0.063) —
Fixed cost parameters	
Affordability index (θ^{in})	−0.007 (0.088) −0.000
PSYPACT (θ_t)	0.217 (0.375) 0.010
Hours (θ_t)	0.305 (0.169) 0.015
Fee (θ_t)	−0.028 (0.108) −0.001
Log likelihood	−690.14
Number of observations	793

Notes: Results are presented on two lines, the first line providing the coefficient and its standard error in parentheses and the second line providing the corresponding marginal effect. Standard errors are calculated using 100 bootstrap replications, while marginal effects are the simulated percent change in the number of LMFTs given a one standard deviation change in the row variable. The dependent variable is total LMFT count per county. LMFT stands for Licensed Marriage and Family Therapists.